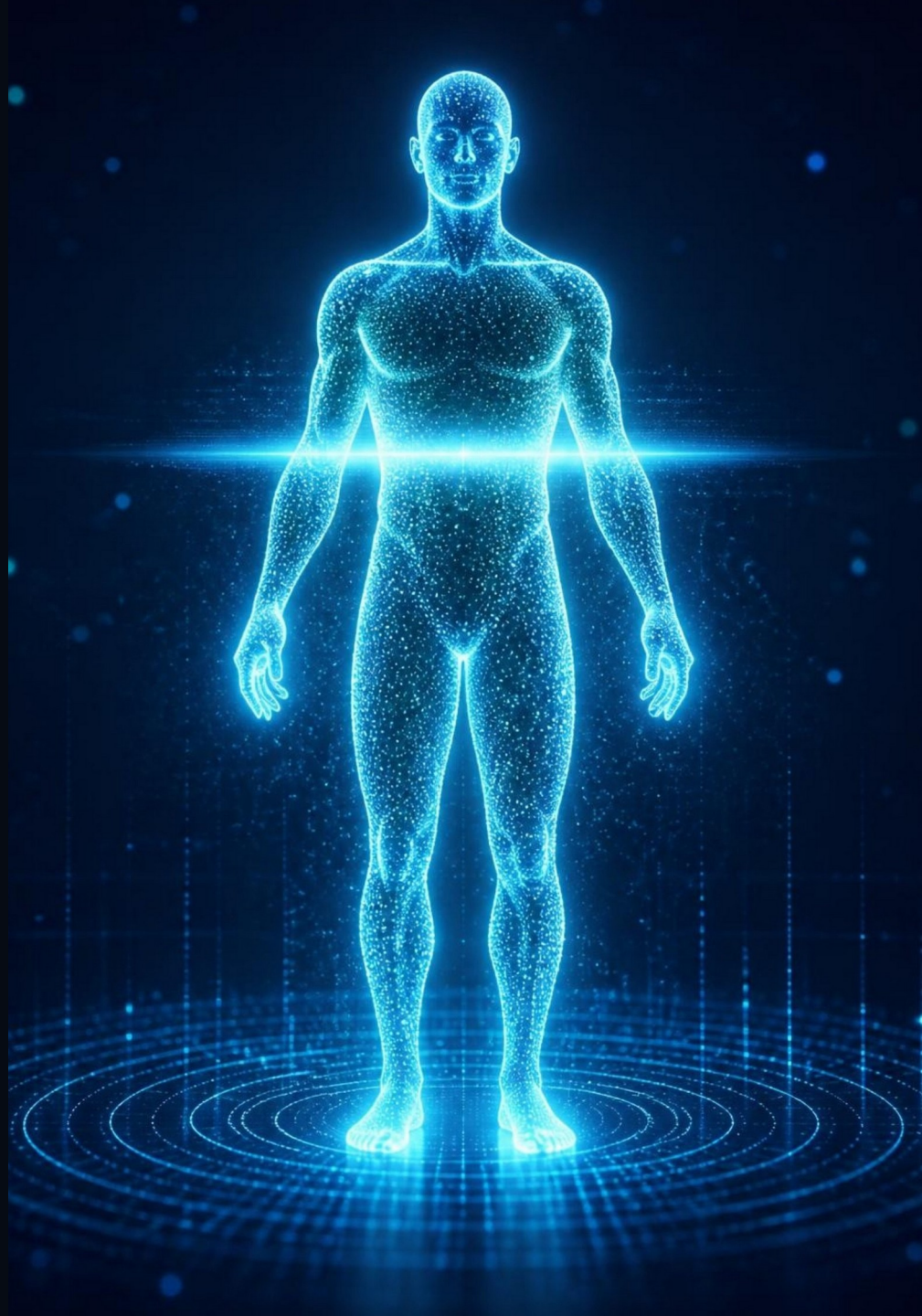
INFINEX 2K26 | APPLIED AI

MedBase Virtual Trials Lab

Rehearse a type 2 diabetes drug trial in your browser, on virtual patients built from real health data, and know how far to trust the result.

Live demo https://zade06079-medbase-virtual-trials-lab.static.hf.space

Team Aegis



THE PROBLEM

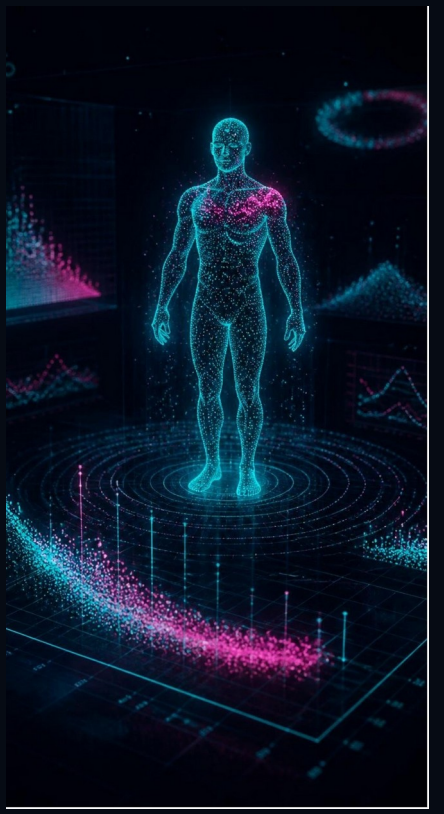
Trials can't be rehearsed, and health AI can't be checked

REHEARSAL GAP

Trials cost years and enormous sums. Students and early researchers have no accessible way to try a study first, and most open diabetes simulators cover type 1 only.

TRUST GAP

Health AI gives confident answers with no way to check them, or to see where it has been wrong before.



MedBase closes both: real patient data, AI that plans and explains, and a simulator that computes every efficacy number.

THE SOLUTION

Ask in plain English, and a real simulator answers

STEP 01

Your question

Ask about a drug, a trial or an idea, in plain English.

STEP 02

Safety guard

Personal and urgent messages stop here, before any AI.

STEP 03

AI copilot

NVIDIA Nemotron plans the experiment, or finds notes by meaning.

STEP 04

Simulator

Computes every efficacy number, right on your device.

STEP 05

Checked answer

Code checks the figures, and the track record sits beside them.

Models propose and explain. The simulator computes.

KEY FEATURES

Six ways to explore a trial

AI copilot

Explains ideas, follows up on "why?", and runs real simulations

Virtual trials

Four real drugs and hypothetical molecules

AI Trial Designer

A neural network proposes a study, and the simulator checks it

Instant neural sliders

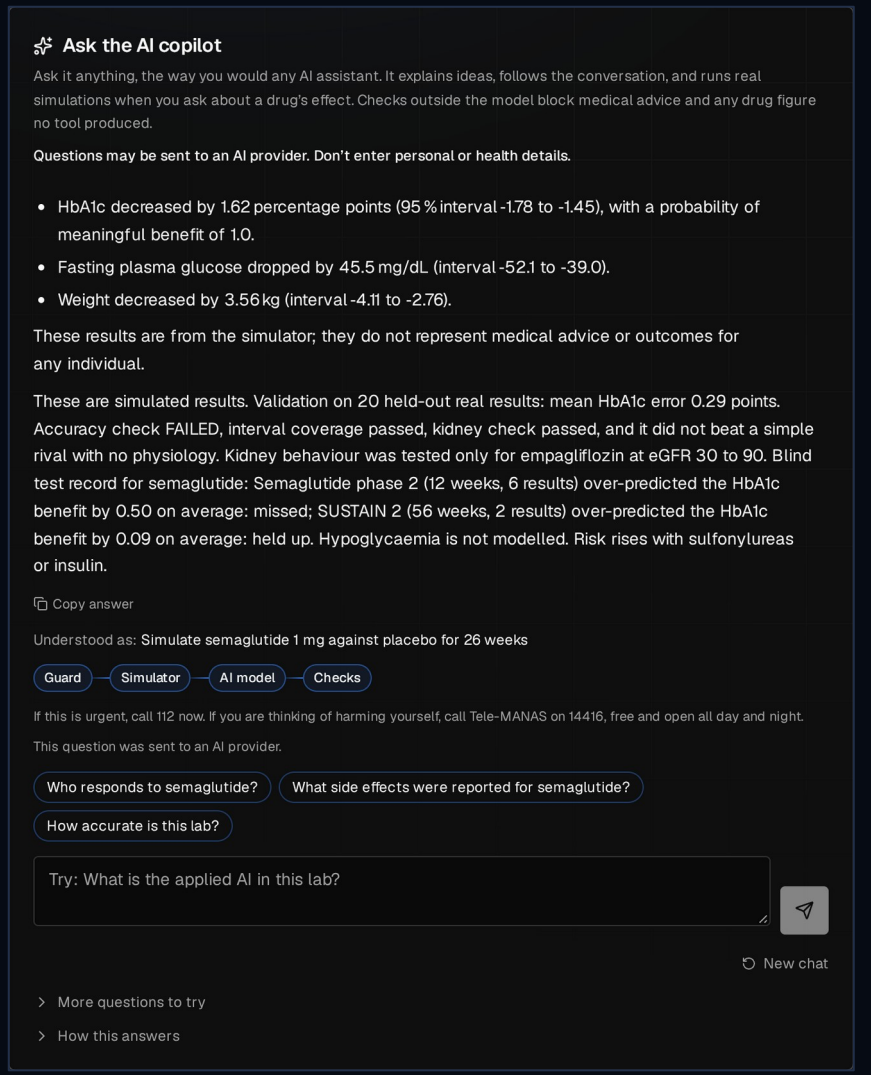
Change dose or duration and see the answer in milliseconds

Vector database

PostgreSQL with pgvector, in the browser, with no server or account

Public blind test

Every frozen prediction beside the real result



TECHNOLOGY STACK

A full AI stack, with the simulator and database in the browser

AI

NVIDIA Nemotron 3.5 on NIM, nemotron-3-embed-1b embeddings, a neural surrogate, Bayesian MCMC with emcee

SCIENCE

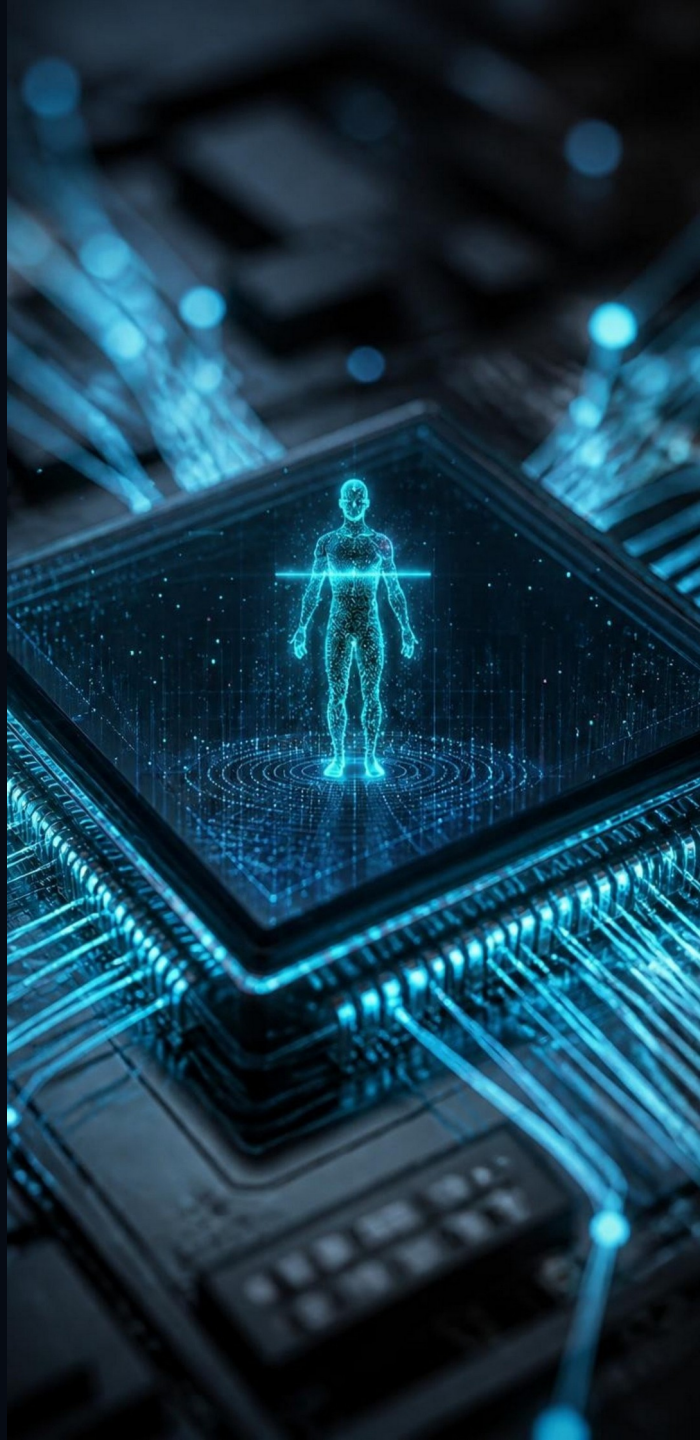
Python, NumPy, SciPy, scikit-learn, statsmodels, NHANES survey data, published trial results

BROWSER

React 19, TypeScript, Vite, Pyodide, PGlite with pgvector 0.8.7, three.js

SAFETY AND HOSTING

Rule-based guard, Vercel key proxy, Hugging Face Spaces, Content Security Policy



INNOVATION

Applied AI you can check

R^2 0.949

Neural surrogate

on 480 simulator runs it never saw

98%

Retrieval

right note in the top 3, on 60 held-out questions

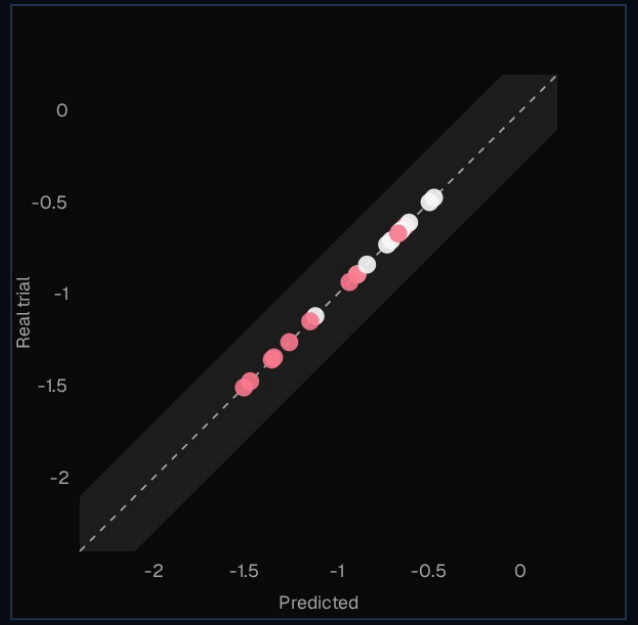
Under 11 ms

Vector database

query measured in the live browser

FROZEN BEFORE THE ANSWERS WERE KNOWN

20 predictions were published before real trial results were opened. 10 landed within 0.30 HbA1c points, the lab passed 2 of 4 pre-registered checks, and every miss stays public. We also attacked our own safety guard and published what got through.



Each dot is a real prediction against the real trial result.

REAL-WORLD IMPACT

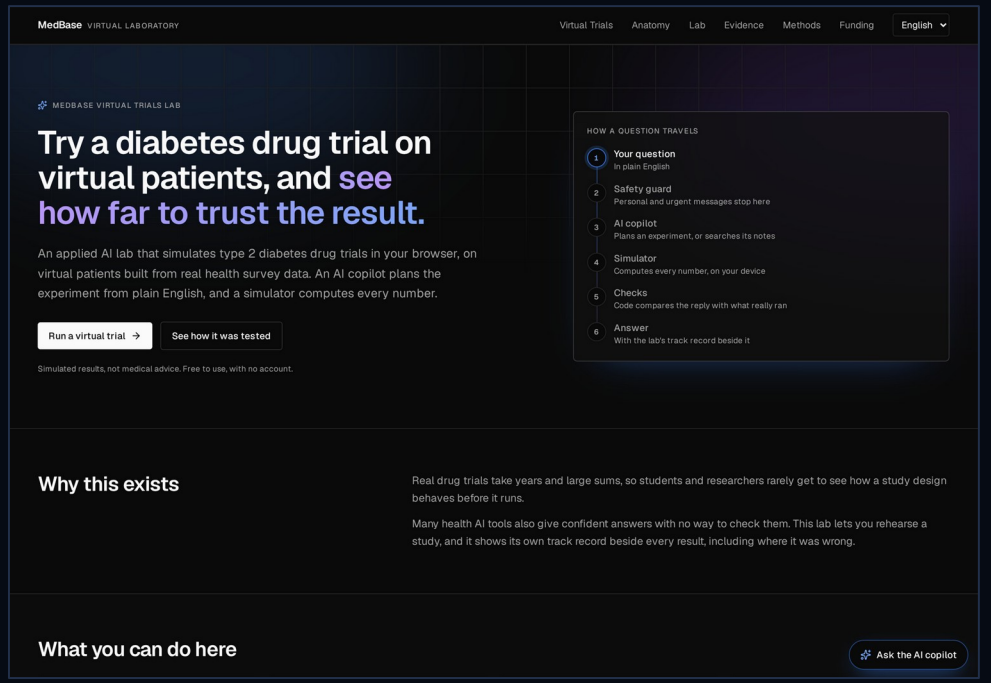
Trial rehearsal for anyone with a laptop

46%

fewer simulator runs than a plain search, across 36 study goals

0

servers to run for the simulator, and no account needed



- Students and early researchers rehearse a study design free, on an ordinary laptop
- Learners see an AI answer's track record right beside it
- Almost no running cost, because each browser does the simulating
- Next: a pilot with a class, and Indian patient data

TRY IT TODAY

Rehearse a trial. See how far to trust it.

Live demo https://zade06079-medbase-virtual-trials-lab.static.hf.space

Code and validation report https://github.com/shreynachaegari-glitch/medbase-virtual-trials-lab

No install. No account. Just ask.

Built on MedBase by Team Aegis